LOGARITHMIC DILATION OF THE JACOBI–RIEMANN KERNEL, STRIP-HOLOMORPHIC SHELL PARTITIONS, STIELTJES DIAGONAL ASYMPTOTICS, AND OFF-DIAGONAL OBSTRUCTIONS

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Abstract. We analyze the logarithmic shell structure of Riemann's Jacobi theta kernel $\Phi(u) = \sum_{m=1}^{\infty} \phi_m(u)$ governing the canonical transverse Wronskian $W(d,t) = \frac{1}{2}\partial_d|\Xi(t - \mathrm{i}d)|^2 = 16\int_0^\infty y\sinh(2dy)I(y,t)\,\mathrm{d}y$. Every shell $\phi_m(u) = m^{-1/2}\varphi_1(u + \frac{1}{2}\ln m)$ is an exact logarithmic translation of $\varphi_1(w)$ whose bilateral Laplace–Mellin transform $\mathcal{G}(s) = \frac{1}{8}s(s - 1)\pi^{-s/2}\Gamma(s/2)$ satisfies $(1 - s)(-s) \equiv s(s - 1)$ on $\mathbb{C}$; combined with the Spira and Saidak–Zvengrowski functional-equation modulus inequality $|\chi(s)| < 1$, this yields strict branch damping across active Riemann–Siegel modes $1 \le n \le N(t) = \lfloor\sqrt{t/2\pi}\rfloor$. We show that previously proposed $\sinh(8z)$ cutoff weights introduce poles at $|\mathrm{Im}(z)| = \frac{\pi}{48} < \frac{\pi}{8}$, whereas both the weight-free even remainder kernel $R_N^{\mathrm{ker}}(z)$ and the logistic partition $w_4(z) = (1 + \mathrm{e}^{-8z})^{-1}$ remain pole-free throughout $|\mathrm{Im}(z)| < \frac{\pi}{8}$. Finally, we prove a cubic-logarithmic Stieltjes diagonal expansion and a sharp $\mathcal{O}(t^{-3/2})$ bound for the boundary pair $(N - 1, N)$, while showing that the $\binom{N(t)}{2}$ off-diagonal pairs sum in L^1 to $\asymp t^{1/2}$ and overwhelm the diagonal pointwise.

1. Introduction and Main Results

In Riemann's memoir [10], the completed zeta function $\xi(s) := \frac{1}{2}s(s - 1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is rotated to the spectral variable $z = t - \mathrm{i}d$ ($s = \frac{1}{2} + \mathrm{i}z = \frac{1}{2} + d + \mathrm{i}t$) via the even entire function $\Xi(z) := \xi(\frac{1}{2} + \mathrm{i}z)$. Using Jacobi's transformation law [6] for $\theta(x) = \sum_{m=-\infty}^{\infty} \mathrm{e}^{-\pi m^2 x}$, $\Xi(z)$ admits the Fourier cosine representation [9, 16]

$$\Xi(z) = 8\int_0^\infty \Phi(u)\cos(2zu)\,\mathrm{d}u = 4\int_{-\infty}^{\infty} \Phi(u)\cos(2zu)\,\mathrm{d}u, \qquad \Phi(u) := \sum_{m=1}^{\infty} \phi_m(u), \tag{1.1}$$

where each shell $\phi_m(u) := (2\pi^2 m^4 \mathrm{e}^{9u} - 3\pi m^2 \mathrm{e}^{5u})\exp(-\pi m^2 \mathrm{e}^{4u})$ is holomorphic on $\mathbb{C}$, while $\Phi(u) = \Phi(-u)$ is even on $\mathbb{R}$ and holomorphic in the open strip $|\mathrm{Im}(u)| < \frac{\pi}{8}$. Writing $\Xi(t - \mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$ for real (d,t), the Cauchy–Riemann equations $R_d = I_t$ and $I_d = -R_t$ identify the canonical transverse Wronskian [13, 7, 4, 1] with the Wigner phase-space integral [17, 5]

$$\boxed{W(d,t) := \frac{1}{2}\frac{\partial}{\partial d}|\Xi(t - \mathrm{i}d)|^2 = R(d,t)I_t(d,t) - I(d,t)R_t(d,t) = 16\int_0^\infty y\sinh(2dy)\,I(y,t)\,\mathrm{d}y,} \tag{1.2}$$

where $I(y,t) := \int_{-\infty}^{\infty} \Phi(\frac{x+y}{2})\Phi(\frac{x-y}{2})\cos(2tx)\,\mathrm{d}x$ is even in y and $\widetilde{W}(d,t) := W(d,t)/d$.

Connecting the continuum representation (1.2) to the finite Riemann–Siegel Dirichlet sum of length $N(t) := \lfloor\sqrt{t/2\pi}\rfloor$ [12, 3] requires care at four distinct analytical points:

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(1) *Archimedean reflection versus collective Dirichlet zeros.* Individual shells generate symmetric pairs $\mathcal{G}(s)m^{-s}+\mathcal{G}(1-s)m^{-(1-s)}$. Clarifying how $|\mathcal{G}(1-s)/\mathcal{G}(s)|=|\chi(s)|<1$ follows from classical results of Spira [14] and Saidak–Zvengrowski [11] separates single-shell Archimedean convexity from collective zero cancellation.
(2) *Strip holomorphy of symmetrized shells.* Naive half-line symmetrization $\phi_m(|u|)$ creates an odd-derivative jump at $u=0$, whereas aggressive double-exponential cutoffs such as $(1+\exp(-2\pi\sinh(8z)))^{-1}$ introduce poles inside $|\mathrm{Im}(z)|<\frac{\pi}{8}$ at $|\mathrm{Im}(z)|=\frac{\pi}{48}$.
(3) *Diagonal asymptotics versus off-diagonal multiplicity.* On the diagonal $m=n$, phase cancellation produces a positive $(\ln t)^3$ Stieltjes expansion and the single boundary pair $(N-1,N)$ is $\mathcal{O}(t^{-3/2})$, yet the $\binom{N(t)}{2}\asymp t$ off-diagonal cross-pairs sum in L^1 to $\asymp t^{1/2}$ and dominate pointwise whenever phases align.
(4) *Contour tail bounds and remainder domains.* Unsplit $\cosh(2(d+\mathrm{i}t)(\xi+\mathrm{i}\eta))$ tail integrals introduce an artificial $\cosh(2t\eta)$ factor, whereas Gabcke's explicit remainder theorem [3] applies strictly on $d=0$ for $t\geq 200$ after subtracting the leading saddle term $\Psi(p)$.

The purpose of this paper is to establish the exact identities governing items (1)–(4) while delineating the analytical obstructions cleanly and without overclaiming.

Theorem 1.1 (Summary of Exact Shell Identities and Analytical Boundaries)**.** *Let $d\in(0,\frac{1}{2}]$, $t\geq 7$, $z=t-\mathrm{i}d$, $s=\frac{1}{2}+d+\mathrm{i}t$, and $N(t)=\lfloor\sqrt{t/2\pi}\rfloor$.*

(i) ***Logarithmic Dilation and Spira Branch Damping.*** *Every shell satisfies $\phi_m(u)=m^{-1/2}\varphi_1(u+\frac{1}{2}\ln m)$ with $\varphi_1(w):=(2\pi^2\mathrm{e}^{9w}-3\pi\mathrm{e}^{5w})\mathrm{e}^{-\pi\mathrm{e}^{4w}}$ and bilateral Laplace–Mellin transform $\mathcal{G}(s)=\frac{1}{8}s(s-1)\pi^{-s/2}\Gamma(s/2)$ ($\mathrm{Re}(s)>-2$). Since $(1-s)(-s)\equiv s(s-1)$ on $\mathbb{C}$, $\rho_\Gamma(d,t):=|\mathcal{G}(1-s)/\mathcal{G}(s)|=|\chi(s)|<\mathrm{e}^{-2d\vartheta'(t)}<1$ for all $d\in(0,\frac{1}{2}]$ and $t\geq 7$ [14, 11], whence $\rho_\Gamma(d,t)n^{2d}<\mathrm{e}^{-2d\omega_n(t)}<1$ for $n=1$ ($t\geq 7$), for every $1\leq n\leq N(t)-1$ ($t\geq 8\pi$), and for $n=N(t)$ whenever $\ln N(t)<\vartheta'(t)$.*

(ii) ***Strip-Holomorphic Symmetrization and Exact Remainder Kernel.*** *The weight $w_{\mathrm{Jac}}(z)=(1+\exp(-2\pi\sinh(8z)))^{-1}$ has poles at $z=\pm\mathrm{i}\frac{\pi}{48}$ inside $|\mathrm{Im}(z)|<\frac{\pi}{8}$. By contrast, the weight-free even remainder kernel $R_N^{\ker}(z):=\Phi(z)-\sum_{m=1}^{N}[\phi_m(z)+\phi_m(-z)]$ and the logistic shells $\Phi_m^{\log}(z):=w_4(z)\phi_m(z)+w_4(-z)\phi_m(-z)$ with $w_4(z):=(1+\mathrm{e}^{-8z})^{-1}$ are holomorphic throughout the full open strip $|\mathrm{Im}(z)|<\frac{\pi}{8}$ and satisfy $(R_N^{\ker})^{(2k+1)}(0)=(\Phi_m^{\log})^{(2k+1)}(0)=0$ for all $k\geq 0$.*

(iii) ***Stieltjes Diagonal Expansion, $(N-1,N)$ Self-Damping, and Off-Diagonal Scaling.*** *With $L:=2\vartheta'(t)=\ln(t/2\pi)+\mathcal{O}(t^{-2})$, the critical-line main-sum diagonal satisfies $\mathcal{D}_{\mathrm{diag}}(0,t):=\sum_{n=1}^{N(t)}\frac{4\omega_n(t)^2}{n}=\frac{1}{6}L^3+\gamma_{\mathrm{E}}L^2-4\gamma_1^{\mathrm{St}}L+4\gamma_2^{\mathrm{St}}+\mathcal{O}(t^{-3/2}\ln^2 t)$, and for $t\geq 8\pi$ ($N=N(t)\geq 2$) the adjacent boundary pair obeys $0<A_{N-1,N}(0,t)<\frac{(3N-2)^2}{[N(N-1)]^{5/2}}<\frac{9}{(N-1)^3}=\mathcal{O}(t^{-3/2})$. However, the total L^1 amplitude of all $\binom{N(t)}{2}\sim\frac{t}{4\pi}$ off-diagonal pairs scales as $\sum_{1\leq m<n\leq N(t)}2A_{m,n}(0,t)=\frac{96}{\sqrt{2\pi}}\,t^{1/2}+\mathcal{O}((\ln t)^2t^{1/4})\gg\mathcal{D}_{\mathrm{diag}}(0,t)$.*

(iv) ***Unshifted Real-Axis Tail Bound and Hyperbolic Sinc Ratio.*** *On the real half-line $\eta=0$, the evanescent tail integral $\mathcal{I}_{\mathrm{tail}}(N,z,0):=\sum_{m=N+1}^{\infty}\int_0^\infty\phi_m(\xi)\cos(2z\xi)\,\mathrm{d}\xi$ satisfies $|\mathcal{I}_{\mathrm{tail}}(N,z,0)|\leq b_{N+1}(d,0)/(1-q_N(0))$ without artificial $\cosh(2t\eta)$ growth, while $\mathrm{sinc}_{\mathrm{hyp}}(dL/2)/\mathrm{sinc}_{\mathrm{hyp}}(dL)\equiv\cosh(dL/2)^{-1}\leq 2(t/2\pi)^{-d/2}$.*

1.1. **Proof overview.** The proofs rest on three analytical ingredients:

(1) *Logarithmic shift-equivariance and tetragamma convexity (Section 2).* Substituting $v=u+\frac{1}{2}\ln m$ reduces the bilateral Mellin integral of ϕ_m to $\mathcal{G}(s)m^{-s}$. Because $(1-s)(-s)=s(s-1)$ identically, $|\mathcal{G}(1-s)/\mathcal{G}(s)|$ reduces to $|\chi(s)|$, whose radial decay on $d\in(0,\frac{1}{2}]$

follows from positivity of the digamma carrier $\mathcal{A}_\Gamma(0,t)=\vartheta'(t)$ above $t_{\mathrm{crit}}\approx 6.289836$ and positive real part of the tetragamma derivative ψ'' [8, 14, 11].

(2) *Pole localization in* $|\mathrm{Im}(z)|<\frac{\pi}{8}$ *(Section 3).* Solving $1+\mathrm{e}^{-8z}=0$ places all poles of $w_4(z)$ at $\mathrm{Im}(z)=\frac{2k+1}{8}\pi$ outside $|\mathrm{Im}(z)|<\frac{\pi}{8}$, whereas $2\pi\sinh(8z)=\pm\mathrm{i}\pi$ places poles of $w_{\mathrm{Jac}}(z)$ at $\mathrm{Im}(z)=\pm\frac{\pi}{48}$. Simultaneously, rewriting the finite bilateral main sum over $[0,\infty)$ isolates the weight-free even remainder kernel $R_N^{\mathrm{ker}}(z)$.

(3) *Euler–Maclaurin boundary vanishing and real-axis incomplete-Gamma integration by parts (Sections 4–5).* Because $f(x)=(\frac{1}{2}L-\ln x)^2/x$ satisfies $f(\mathrm{e}^{L/2})=f'(\mathrm{e}^{L/2})=0$, fractional-part jumps at $N(t)=\lfloor \mathrm{e}^{L/2}\rfloor$ vanish to second order in $\mathcal{D}_{\mathrm{diag}}(0,t)$, while substituting $w=\mathrm{e}^{4\xi}\geq 1$ bounds the $u\geq 0$ tail geometrically.

1.2. **Notation and canonical normalizations.** Throughout, $s=\sigma+\mathrm{i}t$ with $\sigma=\frac{1}{2}+d$, $z=t-\mathrm{i}d$, $\xi(s)=\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)=4\mathcal{G}(s)\zeta(s)$, and $\Xi(z)=\xi(\frac{1}{2}+\mathrm{i}z)$. Note that writing $\widetilde{\Phi}(\tilde{u}):=2\Phi(\tilde{u}/2)$ translates (1.1) into $\Xi(z)=2\int_0^\infty\widetilde{\Phi}(\tilde{u})\cos(z\tilde{u})\,\mathrm{d}\tilde{u}$; we work uniformly in the $(u,2zu)$ variable of (1.1)–(1.2). We set $\vartheta(t):=\mathrm{Im}\ln\Gamma(\frac{1}{4}+\mathrm{i}\frac{t}{2})-\frac{t}{2}\ln\pi$, $\omega_n(t):=\vartheta'(t)-\ln n$, $L:=2\vartheta'(t)$, $N(t):=\lfloor\sqrt{t/2\pi}\rfloor$, and $\mathrm{sinc}_{\mathrm{hyp}}(x):=\frac{\sinh x}{x}$ ($\mathrm{sinc}_{\mathrm{hyp}}(0):=1$). The Stieltjes constants [15] are defined by $\zeta(1+s)=s^{-1}+\sum_{k=0}^\infty\frac{(-1)^k}{\Gamma(k+1)}\gamma_k^{\mathrm{St}}s^k$ with $\gamma_0^{\mathrm{St}}=\gamma_{\mathrm{E}}$.

2. Logarithmic Dilation and the Spira–Saidak–Zvengrowski Modulus Inequality

Proposition 2.1 (Logarithmic Dilation and Bilateral Laplace–Mellin Transform)**.** *Define the generating shell on* $\mathbb{C}$ *by* $\varphi_1(w):=(2\pi^2\mathrm{e}^{9w}-3\pi\mathrm{e}^{5w})\exp(-\pi\mathrm{e}^{4w})$. *Then:*

(i) *For every integer* $m\geq 1$ *and all* $u\in\mathbb{C}$, $\phi_m(u)=m^{-1/2}\varphi_1(u+\frac{1}{2}\ln m)$. *On* $\mathbb{R}$, $\varphi_1(w)$ *has a unique zero at* $w_{\mathrm{node}}=\frac{1}{4}\ln(\frac{3}{2\pi})\approx -0.1848161944$, *a global maximum at* $w_{\mathrm{max}}=\frac{1}{4}\ln(\frac{15+\sqrt{105}}{8\pi})\approx +0.0011334898$, *and a global minimum at* $w_{\mathrm{min}}=\frac{1}{4}\ln(\frac{15-\sqrt{105}}{8\pi})\approx -0.4163462679$.

(ii) *For* $\mathrm{Re}(s)>-2$, *the bilateral Laplace–Mellin integral converges absolutely to*

$$\mathcal{G}(s):=\int_{-\infty}^{\infty}\varphi_1(v)\,\mathrm{e}^{2(s-1/2)v}\,\mathrm{d}v=\frac{1}{8}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right),\tag{2.1}$$

and extends meromorphically to $\mathbb{C}$ *with* $\xi(s)=4\mathcal{G}(s)\zeta(s)$. *Consequently, for every* $m\geq 1$ *and* $z=t-\mathrm{i}d$ *(*$s=\frac{1}{2}+d+\mathrm{i}t$*),*

$$F_{m,\pm}(z):=2\int_{-\infty}^{\infty}\phi_m(u)\cos(2zu)\,\mathrm{d}u=\mathcal{G}(s)m^{-s}+\mathcal{G}(1-s)m^{-(1-s)}.\tag{2.2}$$

Proof. (i) Since $\mathrm{e}^{k(u+\frac{1}{2}\ln m)}=m^{k/2}\mathrm{e}^{ku}$, we have $m^{-1/2}\varphi_1(u+\frac{1}{2}\ln m)=m^{-1/2}(2\pi^2m^{9/2}\mathrm{e}^{9u}-3\pi m^{5/2}\mathrm{e}^{5u})\exp(-\pi m^2\mathrm{e}^{4u})=\phi_m(u)$. Setting $X:=\pi\mathrm{e}^{4w}>0$, $\varphi_1(w)=0\iff 2X=3$, while $\varphi_1'(w)=0\iff 8X^2-30X+15=0\iff X=\frac{15\pm\sqrt{105}}{8}$.

(ii) As $v\to-\infty$, $\varphi_1(v)\mathrm{e}^{(2s-1)v}\sim-3\pi\mathrm{e}^{(2s+4)v}$, which is integrable for $\mathrm{Re}(s)>-2$, while $\exp(-\pi\mathrm{e}^{4v})$ gives super-exponential decay as $v\to+\infty$. Substituting $x=\pi\mathrm{e}^{4v}$ ($\mathrm{d}v=\frac{\mathrm{d}x}{4x}$) gives

$$\mathcal{G}(s)=\frac{\pi^{-s/2}}{4}\int_0^\infty(2x^{s/2+1}-3x^{s/2})\mathrm{e}^{-x}\,\mathrm{d}x=\frac{\pi^{-s/2}}{4}\left[2\Gamma\left(\frac{s}{2}+2\right)-3\Gamma\left(\frac{s}{2}+1\right)\right],$$

which simplifies via $\Gamma(w+1)=w\Gamma(w)$ to $\frac{1}{4}\pi^{-s/2}\frac{s}{2}(\frac{s}{2}-\frac{1}{2})\Gamma(\frac{s}{2})=\frac{1}{8}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})$. Writing $2\cos(2zu)=\mathrm{e}^{2(s-1/2)u}+\mathrm{e}^{2((1-s)-1/2)u}$ and substituting $v=u+\frac{1}{2}\ln m$ yields (2.2). □

We now connect (2.2) to the classical functional-equation factor $\chi(s) := \pi^{s-1/2}\Gamma(\frac{1-s}{2})/\Gamma(\frac{s}{2})$ studied by Spira [14] and Saidak and Zvengrowski [11].

Proposition 2.2 (Spira–Saidak–Zvengrowski Branch Damping)**.** *Let $s = \frac{1}{2} + d + \mathrm{i}t$ ($d > 0$, $t \geq 7$). Set $\mathcal{A}_\Gamma(d,t) := \frac{1}{2}\mathrm{Re}\,\psi(\frac{1}{4} + \frac{d}{2} + \mathrm{i}\frac{t}{2}) - \frac{1}{2}\ln\pi$ and $\mathcal{A}_\Gamma^{\mathrm{sym}}(d,t) := \frac{1}{2}[\mathcal{A}_\Gamma(d,t) + \mathcal{A}_\Gamma(-d,t)]$. Then:*

(i) *By elementary algebra, $(1-s)(-s) = s^2 - s \equiv s(s-1)$ holds identically for all $s \in \mathbb{C}$. Hence the polynomial prefactor cancels identically in $\mathcal{G}(1-s)/\mathcal{G}(s) = \chi(s)$, and*

$$\rho_\Gamma(d,t) := \left|\frac{\mathcal{G}(1-s)}{\mathcal{G}(s)}\right| = |\chi(s)| = \pi^d \left|\frac{\Gamma(\frac{1}{4} - \frac{d}{2} + \mathrm{i}\frac{t}{2})}{\Gamma(\frac{1}{4} + \frac{d}{2} + \mathrm{i}\frac{t}{2})}\right| = \exp\left(-2\int_0^d \mathcal{A}_\Gamma^{\mathrm{sym}}(u,t)\,\mathrm{d}u\right). \tag{2.3}$$

(ii) *For all $t > t_{\mathrm{crit}} = 6.28983598883690277967\ldots$ (in particular $t \geq 7$) and $d \in (0, \frac{1}{2}]$, $\mathcal{A}_\Gamma^{\mathrm{sym}}(u,t) > \mathcal{A}_\Gamma(0,t) = \vartheta'(t) > 0$ for $u \in (0,d]$, so $\rho_\Gamma(d,t) < \mathrm{e}^{-2d\vartheta'(t)} < 1$ [14, 11].*

(iii) *For $d \in (0, \frac{1}{2}]$, every mode $n = 1$ ($t \geq 7$) and $1 \leq n \leq N(t) - 1$ ($t \geq 8\pi$) satisfies $\omega_n(t) = \vartheta'(t) - \ln n > 0$ and the strict branch-damping inequality*

$$\rho_\Gamma(d,t)\, n^{-1/2+d} < \mathrm{e}^{-2d\omega_n(t)}\, n^{-1/2-d} < n^{-1/2-d}, \tag{2.4}$$

which also holds at $n = N(t)$ whenever $\ln N(t) < \vartheta'(t)$ (outside the transition micro-intervals $2\pi m^2 \leq t < t_m^*$ where $\vartheta'(t_m^*) = \ln m$ and $\omega_m(t) \in (-\frac{1}{24t^2}, 0]$).

Proof. (i) Dividing $\mathcal{G}(1-s) = \frac{1}{8}(1-s)(-s)\pi^{-(1-s)/2}\Gamma(\frac{1-s}{2})$ by $\mathcal{G}(s)$ and using $(1-s)(-s) = s(s-1)$ gives $\mathcal{G}(1-s)/\mathcal{G}(s) = \chi(s)$. Since $\partial_u \ln|\Gamma(\frac{1}{4} \pm \frac{u}{2} + \mathrm{i}\frac{t}{2})| = \pm\frac{1}{2}\mathrm{Re}\,\psi(\frac{1}{4} \pm \frac{u}{2} + \mathrm{i}\frac{t}{2})$ and $\ln|\Gamma(\bar{w})| = \ln|\Gamma(w)|$, integrating $\partial_u \ln\rho_\Gamma(u,t) = -2\mathcal{A}_\Gamma^{\mathrm{sym}}(u,t)$ from 0 to d with $\rho_\Gamma(0,t) = 1$ yields (2.3).

(ii) By Stirling's series with Olver's remainder [8], $\vartheta'(t) = \frac{1}{2}\ln(\frac{t}{2\pi}) + \mathcal{O}(t^{-2})$ crosses zero uniquely at $t_{\mathrm{crit}} \approx 6.289836 < 7$. Moreover, $\partial_u^2\mathcal{A}_\Gamma^{\mathrm{sym}}(u,t) = \frac{1}{16}[\mathrm{Re}\,\psi''(\frac{1}{4} + \frac{u}{2} + \mathrm{i}\frac{t}{2}) + \mathrm{Re}\,\psi''(\frac{1}{4} - \frac{u}{2} + \mathrm{i}\frac{t}{2})]$. For $w = x + \mathrm{i}y$ with $x \in [0, \frac{1}{2}]$ and $y = \frac{t}{2} \geq \frac{7}{2}$, set $w_1 := w + 1 = x_1 + \mathrm{i}y$ ($x_1 \in [1, \frac{3}{2}]$), so $\psi''(w) = \psi''(w_1) - 2w^{-3}$. Since $0 \leq x < x_1 \leq \frac{3}{2} < \sqrt{3}\,y$, both $\mathrm{Re}(-2w^{-3}) = 2x(3y^2 - x^2)|w|^{-6} \geq 0$ and $\mathrm{Re}(-w_1^{-3}) > 0$. Applying second-order Euler–Maclaurin summation to $\psi''(w_1) = -2\sum_{k=0}^\infty (w_1+k)^{-3}$ gives $\psi''(w_1) = -w_1^{-2} - w_1^{-3} + R_2(w_1)$, where $R_2(w_1) = -12\int_0^\infty \{\tau\}(1-\{\tau\})(w_1+\tau)^{-5}\,\mathrm{d}\tau$ obeys $|R_2(w_1)| \leq 3\int_0^\infty ((x_1+\tau)^2 + y^2)^{-5/2}\,\mathrm{d}\tau < 3y^{-4}\int_0^\infty (s^2+1)^{-5/2}\,\mathrm{d}s = 2y^{-4}$. Since $\mathrm{Re}(-w_1^{-2}) = (y^2 - x_1^2)(x_1^2 + y^2)^{-2} \geq (y^2 - \frac{9}{4})(y^2 + \frac{9}{4})^{-2} > 2y^{-4}$ for all $y \geq \frac{7}{2}$ (as $y^2(y^4 - \frac{17}{4}y^2 - 9) - \frac{81}{8} \geq \frac{8641}{8} > 0$), we obtain $\mathrm{Re}\,\psi''(w) > 0$, hence $\partial_u^2\mathcal{A}_\Gamma^{\mathrm{sym}}(u,t) > 0$. Because $\partial_u\mathcal{A}_\Gamma^{\mathrm{sym}}(0,t) = 0$, $\mathcal{A}_\Gamma^{\mathrm{sym}}(u,t) > \vartheta'(t) > 0$ on $(0,d]$, proving $\rho_\Gamma(d,t) < \mathrm{e}^{-2d\vartheta'(t)} < 1$.

(iii) For $t \geq 7$, $\omega_1(t) = \vartheta'(t) > 0$. For $t \geq 8\pi$ ($N = N(t) \geq 2$), Olver's remainder bound [8, Ch. 8] gives $\vartheta'(t) > \frac{1}{2}\ln(\frac{t}{2\pi}) - \frac{1}{24t^2} \geq \ln N - \frac{1}{24t^2}$. Thus for $1 \leq n \leq N-1$, $\omega_n(t) \geq \omega_{N-1}(t) > \ln(\frac{N}{N-1}) - \frac{1}{24t^2} > \frac{1}{N} - \frac{1}{96\pi^2N^4} > 0$, and at $n = N$ the inequality $\omega_N(t) > 0$ holds whenever $\ln N < \vartheta'(t)$, giving (2.4).

□

Remark 2.3 (Single-shell positivity versus Davenport–Heilbronn counterexamples). For $m = 1$, $F_{1,\pm}(z) = \mathcal{G}(s) + \mathcal{G}(1-s)$ satisfies $|F_{1,\pm}(t - \mathrm{i}d)| \geq |\mathcal{G}(s)|(1 - \rho_\Gamma(d,t)) > 0$ and has positive self-Wronskian $W_{1,1}(d,t) := \frac{1}{2}\partial_d|F_{1,\pm}(t - \mathrm{i}d)|^2 > 0$ on $(0, \frac{1}{2}] \times [7, \infty)$ (Table 1). However, $F_{1,\pm}$ represents only the single mode $n = 1$ of $\zeta(s)$. Any Dirichlet series sharing the same Gamma factor $\mathcal{G}(s)$—such as the Davenport–Heilbronn function [2], which lacks an Euler product and has infinitely many zeros in $\frac{1}{2} < \sigma < 1$—enjoys the identical inequality $\rho_\Gamma(d,t) < 1$. Thus Proposition 2.2 quantifies the Archimedean branch asymmetry of each individual mode, not collective non-vanishing.

TABLE 1. 50-decimal `mpmath` benchmarks of the Spira–Saidak–Zvengrowski modulus ratio $\rho_\Gamma(d,t) = |\chi(s)| < 1$, horizon branch ratio $\rho_\Gamma(d,t)N(t)^{2d} < 1$, and normalized single-shell self-Wronskian $W_{1,1}(d,t)/|\mathcal{G}(s)|^2 > 0$.

d	Height t	$N(t)$	$\rho_\Gamma(d,t)$	$\rho_\Gamma(d,t)\,N(t)^{2d}$	$W_{1,1}(d,t)/\lvert\mathcal{G}(s)\rvert^2$
0.01	7.000000	1	0.99892877	0.99892877	0.000367
0.05	14.134725	1	0.96028272	0.96028272	0.032651
0.25	14.134725	1	0.81656380	0.81656380	0.139841
0.50	25.010858	1	0.50121683	0.50121683	0.524149
0.10	100.00000	3	0.75826002	0.94458797	0.588172
0.25	1000.0000	12	0.28154315	0.97529409	2.333957

3. Wigner Shift Levers, Pole Obstruction of $\sinh(8z)$ Weights, and Strip-Holomorphic Shells

Rotating $(u,v) \in \mathbb{R}^2$ to Wigner coordinates $x = u+v$, $y = u-v$ ($\mathrm{d}u\,\mathrm{d}v = \frac{1}{2}\mathrm{d}x\,\mathrm{d}y$) in $|\Xi(t-\mathrm{i}d)|^2 = 16\iint_{\mathbb{R}^2}\Phi(u)\Phi(v)\cos(2(t-\mathrm{i}d)u)\cos(2(t+\mathrm{i}d)v)\,\mathrm{d}u\,\mathrm{d}v$ and using $\Phi(-u) = \Phi(u)$ yields $|\Xi(t-\mathrm{i}d)|^2 = 16\int_0^\infty \cosh(2dy)I(y,t)\,\mathrm{d}y$, whose $\frac{1}{2}\partial_d$ derivative gives the canonical formula (1.2).

Theorem 3.1 (Wigner Shift Lever, Pole Obstruction of w_{Jac}, and Strip-Holomorphic Replacements)**.** *Let $z \in \mathbb{C}$ and $N \geq 1$.*

(i) ***Wigner Phase-Space Shift and Canonical Lever.*** *For any shells $m, n \geq 1$,*

$$\phi_m\!\left(\frac{x+y}{2}\right)\phi_n\!\left(\frac{x-y}{2}\right) = \frac{1}{\sqrt{mn}}\,\varphi_1\!\left(\frac{(x-x_{m,n})+(y-y_{m,n})}{2}\right)\varphi_1\!\left(\frac{(x-x_{m,n})-(y-y_{m,n})}{2}\right), \tag{3.1}$$

centered at $(x_{m,n}, y_{m,n}) := (-\frac{1}{2}\ln(mn), \frac{1}{2}\ln(n/m))$, where the canonical Wronskian lever at $y = y_{m,n}$ matches the Dirichlet exponent d:

$$\mathcal{L}_{m,n}(d) := y_{m,n}\sinh(2dy_{m,n}) = \frac{1}{4}\ln\!\left(\frac{n}{m}\right)\left[\left(\frac{n}{m}\right)^d - \left(\frac{m}{n}\right)^d\right] > 0 \qquad (m \neq n,\ d > 0). \tag{3.2}$$

(ii) ***Pole Obstruction of Double-Exponential*** $\sinh(8z)$ ***Cutoffs.*** *Although each individual shell has a non-zero odd derivative at the origin ($\phi_1'(0) = -\sum_{m=2}^\infty \phi_m'(0) = (18\pi^2 - 15\pi - 8\pi^3 + 12\pi^2)\mathrm{e}^{-\pi} = +0.03949876526774993468\ldots \neq 0$), symmetrizing via $w_{\mathrm{Jac}}(z) := (1+\exp(-2\pi\sinh(8z)))^{-1}$ fails to preserve strip holomorphy: $w_{\mathrm{Jac}}(z)$ has poles wherever $\sinh(8z) = \mathrm{i}(k+\frac{1}{2})$ ($k \in \mathbb{Z}$), including $z = \pm\mathrm{i}\frac{\pi}{48}$ ($|\mathrm{Im}(z)| = \frac{\pi}{48} \approx 0.065450 < \frac{\pi}{8} \approx 0.392699$) and infinitely many poles along the horizontal lines $\mathrm{Im}(z) = \pm\frac{\pi}{16}$ where $\cosh(8\,\mathrm{Re}(z)) = k + \frac{1}{2}$ ($k \geq 1$).*

(iii) ***Weight-Free Even Remainder Kernel.*** *Separating the N-term bilateral main sum from (1.1) yields without any cutoff weight the exact identity*

$$\Xi(z) = 4\sum_{m=1}^{N}\left[\mathcal{G}(s)m^{-s} + \mathcal{G}(1-s)m^{-(1-s)}\right] + 8\int_0^\infty R_N^{\mathrm{ker}}(u)\cos(2zu)\,\mathrm{d}u, \tag{3.3}$$

where $R_N^{\mathrm{ker}}(u) := \Phi(u) - \sum_{m=1}^N[\phi_m(u) + \phi_m(-u)]$ is even on $\mathbb{R}$, holomorphic throughout the full open strip $|\mathrm{Im}(u)| < \frac{\pi}{8}$, and satisfies $(R_N^{\mathrm{ker}})^{(2k+1)}(0) = 0$ for all $k \geq 0$.

(iv) ***Strip-Holomorphic Logistic Shell Partition.*** *Define $w_4(z) := (1+\mathrm{e}^{-8z})^{-1}$ and $\Phi_m^{\log}(z) := w_4(z)\phi_m(z) + w_4(-z)\phi_m(-z)$. Then $w_4(z) + w_4(-z) \equiv 1$, the poles of $w_4(z)$ lie at $z = \mathrm{i}\frac{2k+1}{8}\pi$ ($k \in \mathbb{Z}$) strictly outside the open strip $|\mathrm{Im}(z)| < \frac{\pi}{8}$, every shell $\Phi_m^{\log}(z)$*

is even and holomorphic on $|\mathrm{Im}(z)| < \frac{\pi}{8}$ *with* $(\Phi_m^{\log})^{(2k+1)}(0) = 0$ *for all* $k \geq 0$, *and* $\sum_{m=1}^{\infty} \Phi_m^{\log}(z) \equiv \Phi(z)$ *on* $|\mathrm{Im}(z)| < \frac{\pi}{8}$.

Proof. (i) Applying Proposition 2.1(i) to $\frac{x+y}{2} + \frac{1}{2}\ln m = \frac{(x-x_{m,n})+(y-y_{m,n})}{2}$ and $\frac{x-y}{2} + \frac{1}{2}\ln n = \frac{(x-x_{m,n})-(y-y_{m,n})}{2}$ yields (3.1); substituting $y_{m,n} = \frac{1}{2}\ln(n/m)$ into $y_{m,n}\sinh(2dy_{m,n})$ gives (3.2).

(ii) Differentiating $\phi_1(u)$ at $u = 0$ gives $\phi_1'(0) = (30\pi^2 - 15\pi - 8\pi^3)\mathrm{e}^{-\pi} \approx +0.0394987653$, and $\Phi'(0) = 0$ implies $\sum_{m=2}^{\infty}\phi_m'(0) = -\phi_1'(0)$. The denominator of $w_{\mathrm{Jac}}(z)$ vanishes iff $2\pi\sinh(8z) = \mathrm{i}(2k+1)\pi \iff \sinh(8\xi)\cos(8\eta) + \mathrm{i}\cosh(8\xi)\sin(8\eta) = \mathrm{i}(k+\frac{1}{2})$ $(z = \xi + \mathrm{i}\eta)$. Setting $\xi = 0$, $k = 0$ gives $\sin(8\eta) = \frac{1}{2}$, hence $\eta = \frac{\pi}{48} < \frac{\pi}{8}$; setting $\eta = \frac{\pi}{16}$ gives $\cos(8\eta) = 0$ and $\cosh(8\xi) = k + \frac{1}{2}$ for every integer $k \geq 1$.

(iii) By (2.2) and evenness of $\cos(2zu)$,

$$\begin{aligned} 4\sum_{m=1}^{N}\left[\mathcal{G}(s)m^{-s} + \mathcal{G}(1-s)m^{-(1-s)}\right] &= 8\sum_{m=1}^{N}\int_{-\infty}^{\infty}\phi_m(u)\cos(2zu)\,\mathrm{d}u \\ &= 8\sum_{m=1}^{N}\int_{0}^{\infty}\left[\phi_m(u)+\phi_m(-u)\right]\cos(2zu)\,\mathrm{d}u. \end{aligned}$$

Subtracting this from $\Xi(z) = 8\int_0^{\infty}\Phi(u)\cos(2zu)\,\mathrm{d}u$ yields (3.3). Since $\Phi(-u) = \Phi(u)$ and $\phi_m(u) + \phi_m(-u)$ are both even and holomorphic on $|\mathrm{Im}(u)| < \frac{\pi}{8}$, so is $R_N^{\ker}(u)$, and all its odd derivatives vanish at $u = 0$.

(iv) We have $w_4(z) + w_4(-z) = (1+\mathrm{e}^{-8z})^{-1} + (1+\mathrm{e}^{8z})^{-1} = 1$, and $1 + \mathrm{e}^{-8z} = 0 \iff 8z = \mathrm{i}(2k+1)\pi \iff \mathrm{Im}(z) = \frac{2k+1}{8}\pi$, so $|\mathrm{Im}(z)| \geq \frac{\pi}{8}$ at every pole. For any fixed z with $|\mathrm{Im}(z)| < \frac{\pi}{8}$, $\cos(4\,\mathrm{Im}(\pm z)) = \cos(4\,\mathrm{Im}(z)) > 0$, so $\phi_m(\pm z)$ decays super-exponentially in m^2 and $\sum_{m=1}^{\infty}\Phi_m^{\log}(z) = w_4(z)\Phi(z) + w_4(-z)\Phi(-z) = \Phi(z)$ (noting that as $\xi = \mathrm{Re}(z) \to +\infty$, $|w_4(-z)\phi_m(-z)| \asymp m^2\mathrm{e}^{-13\xi}$ decays integrably against $\mathrm{e}^{2d\xi}$ for all $d \leq \frac{1}{2}$, while at $\xi = \frac{1}{2}\ln m$ the reflected peak has algebraic size $w_4(-\frac{1}{2}\ln m)\phi_m(-\frac{1}{2}\ln m) = \mathcal{O}(m^{-9/2})$). □

4. Stieltjes Diagonal Asymptotics, $(N-1, N)$ Self-Damping, and Off-Diagonal Obstructions

Factoring out $16|\mathcal{G}(\frac{1}{2}+\mathrm{i}t)|^2$ from the $N(t)$-term main sum in (3.3), the leading-order parity-reduced Wronskian sum $\widehat{W}_{\mathrm{lead}}(d,t) = \mathcal{D}_{\mathrm{diag}}(d,t) + \mathcal{O}_{\mathrm{off}}(d,t)$ has diagonal part

$$\mathcal{D}_{\mathrm{diag}}(d,t) := \sum_{n=1}^{N(t)}\frac{4\,\omega_n(t)^2}{n}\,\mathrm{sinc}_{\mathrm{hyp}}(2d\,\omega_n(t)) > 0, \qquad \omega_n(t) = \vartheta'(t) - \ln n = \frac{1}{2}L - \ln n, \quad (4.1)$$

because on $m = n$ the oscillatory phases satisfy $\sin^2\phi_n(t) + \cos^2\phi_n(t) \equiv 1$ $(\phi_n(t) := \vartheta(t) - t\ln n)$ and the difference frequency vanishes.

Theorem 4.1 (Cubic-Logarithmic Stieltjes Expansion of $\mathcal{D}_{\mathrm{diag}}(0,t)$)**.** *Let* $t \geq 14$, $L := 2\vartheta'(t) = \ln(\frac{t}{2\pi}) + \mathcal{O}(t^{-2})$, *and* $N(t) = \lfloor\sqrt{t/2\pi}\rfloor$. *On the critical line* $d = 0$, *the non-oscillatory diagonal sum obeys the unconditional Euler–Maclaurin expansion*

$$\mathcal{D}_{\mathrm{diag}}(0,t) = \sum_{n=1}^{N(t)}\frac{(L-2\ln n)^2}{n} = \frac{1}{6}L^3 + \gamma_{\mathrm{E}}L^2 - 4\gamma_1^{\mathrm{St}}L + 4\gamma_2^{\mathrm{St}} + \mathcal{O}(t^{-3/2}\ln^2 t), \quad (4.2)$$

where $\gamma_{\mathrm{E}} \approx 0.5772156649$, $\gamma_1^{\mathrm{St}} \approx -0.07281584548$, *and* $\gamma_2^{\mathrm{St}} \approx -0.00969036319$, *and for* $d \in (0, \frac{1}{2}]$ *the integral main term is* $\frac{dL\cosh(dL) - \sinh(dL)}{2d^3} + \mathcal{O}(\mathrm{e}^{dL}L)$.

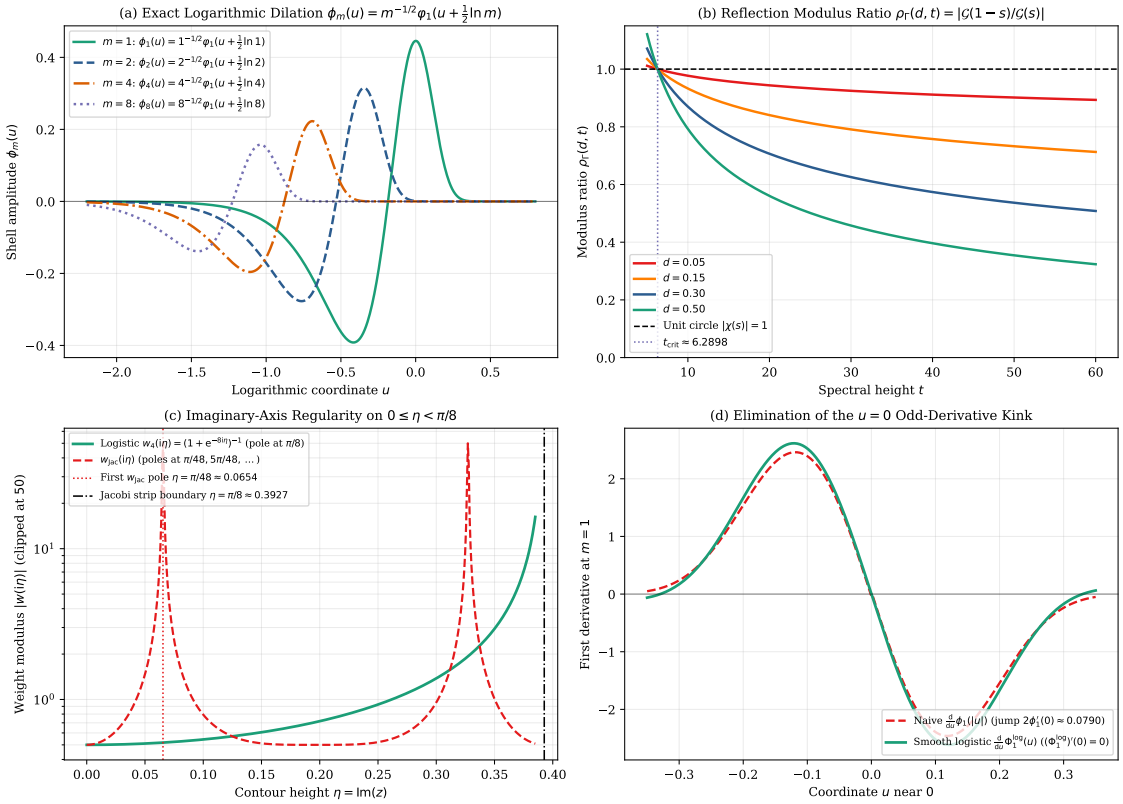


FIGURE 1. **(a)** Exact logarithmic shell dilation $\phi_m(u) = m^{-1/2}\varphi_1(u + \frac{1}{2}\ln m)$. **(b)** Spira–Saidak–Zvengrowski modulus ratio $\rho_\Gamma(d, t) = |\chi(s)| < 1$ above $t_{\text{crit}} \approx 6.2898$. **(c)** Imaginary-axis regularity showing the poles of $w_{\text{Jac}}(i\eta)$ at $\eta = \frac{\pi}{48}, \frac{5\pi}{48}, \ldots$ inside $[0, \frac{\pi}{8})$ versus the pole-free logistic weight $w_4(i\eta) = (1+e^{-8i\eta})^{-1}$. **(d)** Elimination of the $u = 0$ derivative jump $2\phi_1'(0) \approx 0.0790$ by $\Phi_1^{\log}(u)$.

Proof. Expand $\mathcal{D}_{\text{diag}}(0, t) = L^2 H_N - 4LH_N^{(1)} + 4H_N^{(2)}$, where $H_N^{(r)} := \sum_{n=1}^{N} \frac{(\ln n)^r}{n}$ $(N = N(t))$. Writing $N = x_t - \delta_t$ with $x_t := e^{L/2} = \sqrt{t/2\pi} + \mathcal{O}(t^{-3/2})$ and $\delta_t \in [0, 1) + \mathcal{O}(t^{-3/2})$, Stieltjes's generalized Euler–Maclaurin formula [15, 8] gives

$$H_N^{(r)} = \frac{(\ln N)^{r+1}}{r+1} + \gamma_r^{\text{St}} + \frac{(\ln N)^r}{2N} - \frac{r(\ln N)^{r-1} - (\ln N)^r}{12N^2} + \mathcal{O}(N^{-3}(\ln N)^r).$$

Substituting $\ln N = \frac{1}{2}L - \frac{\delta_t}{x_t} - \frac{\delta_t^2}{2x_t^2} + \mathcal{O}(x_t^{-3})$ into the polynomial combination $G(y) := L^2 y - 2Ly^2 + \frac{4}{3}y^3$ at $y = \ln N$, the derivatives at $y_0 = \frac{1}{2}L$ satisfy $G(y_0) = \frac{1}{6}L^3$, $G'(y_0) = L^2 - 4L(\frac{1}{2}L) + 4(\frac{1}{2}L)^2 \equiv 0$, and $G''(y_0) = -4L + 8(\frac{1}{2}L) \equiv 0$. Furthermore, at $n = N$ the boundary integrand $f(x) = (L - 2\ln x)^2/x$ satisfies $f(x_t) = 0$ and $f'(x_t) = 0$, so both the $\mathcal{O}(x_t^{-1})$ and $\mathcal{O}(x_t^{-2})$ sawtooth jump terms cancel identically, leaving $\frac{1}{6}L^3 + \gamma_{\text{E}}L^2 - 4\gamma_1^{\text{St}}L + 4\gamma_2^{\text{St}} + \mathcal{O}(x_t^{-3}\ln^2 x_t) = \cdots + \mathcal{O}(t^{-3/2}\ln^2 t)$. For $d > 0$, substituting $v = L - 2\ln x$ in $\int_1^{e^{L/2}} \frac{2(L-2\ln x)}{dx} \sinh(d(L - 2\ln x))\,\mathrm{d}x = \frac{1}{d}\int_0^L v\sinh(dv)\,\mathrm{d}v$ gives $\frac{dL\cosh(dL) - \sinh(dL)}{2d^3}$. □

We next establish the sharp bound for the adjacent boundary cross-pair $(N-1, N)$ together with the global off-diagonal scaling obstruction.

Proposition 4.2 (Sharp $(N-1,N)$ Self-Damping versus $\mathcal{O}(t^{1/2})$ Cross-Pair Multiplicity)**.** *Let $t \geq 8\pi$ so that $N = N(t) = \lfloor \sqrt{t/2\pi} \rfloor \geq 2$, and for $1 \leq m < n \leq N$ denote the critical-line slow difference-frequency cross-pair amplitude by $A_{m,n}(0,t) := \frac{(\omega_m(t)+\omega_n(t))^2}{\sqrt{mn}}$ (contributing $2A_{m,n}(0,t)\cos((t\ln\frac{n}{m}))$ to $\widehat{W}_{\mathrm{lead}}(0,t)$ up to $\mathcal{O}(t^{-2})$). Then:*

(i) ***Sharp $\mathcal{O}(t^{-3/2})$ Self-Damping of the Single Boundary Pair*** $(N-1,N)$**.** *For all $t \geq 8\pi$ ($N = N(t) \geq 2$), using $\vartheta'(t) < \frac{1}{2}\ln(\frac{t}{2\pi}) = \ln(N+\delta_t)$ with $\delta_t := \sqrt{t/2\pi} - N \in [0,1)$, we have*

$$0 < A_{N-1,N}(0,t) < \frac{(3N-2)^2}{\big[N(N-1)\big]^{5/2}} < \frac{9}{(N-1)^3} = \mathcal{O}(t^{-3/2}). \tag{4.3}$$

(ii) ***Pointwise Off-Diagonal Obstruction.*** *Across all $\binom{N}{2} = \frac{N(N-1)}{2} \sim \frac{t}{4\pi}$ off-diagonal pairs $1 \leq m < n \leq N$, the total L^1 cross-amplitude sum scales as*

$$\mathcal{S}^{L^1}_{\mathrm{off}}(0,t) := \sum_{1\leq m<n\leq N(t)} 2A_{m,n}(0,t) = \frac{96}{\sqrt{2\pi}}\,t^{1/2} + \mathcal{O}\big((\ln t)^2\,t^{1/4}\big) \gg \frac{1}{6}(\ln t)^3 \sim \mathcal{D}_{\mathrm{diag}}(0,t). \tag{4.4}$$

Consequently, diagonal positivity $\mathcal{D}_{\mathrm{diag}}(d,t) > 0$ never implies pointwise positivity of $\widehat{W}_{\mathrm{lead}}(d,t)$: on $d = 0$, $\widetilde{W}(0,t) = (\Xi'(t))^2 - \Xi(t)\Xi''(t)$ can only be controlled pointwise when phases cancel, while Fejér tent averaging eliminating all cross-frequencies $\geq \Omega_{\min}(t) = \ln(\frac{N}{N-1}) \sim \sqrt{2\pi/t}$ requires window half-width $\Delta T \geq \frac{2\pi}{\Omega_{\min}(t)} \sim \sqrt{2\pi^3 t}$, spanning $\asymp \sqrt{t}\ln t$ critical-line zeros.

Proof. (i) By Stirling's expansion $\operatorname{Re}\psi(\frac{1}{4}+\mathrm{i}\frac{t}{2}) = \ln|\frac{1}{4}+\mathrm{i}\frac{t}{2}| - \frac{1}{8|w|^2} - \frac{\cos(2\arg w)}{12|w|^2} + \dots$, for all $t \geq 8\pi$ ($N \geq 2$) we have $\ln N - \frac{1}{24t^2} < \vartheta'(t) < \frac{1}{2}\ln(\frac{t}{2\pi}) = \ln(N+\delta_t)$ with $\delta_t \in [0,1)$. Even when $\omega_N(t) \in (-\frac{1}{24t^2}, 0]$ near $t = 2\pi N^2$, $\omega_{N-1}(t) + \omega_N(t) > \ln(\frac{N}{N-1}) - \frac{1}{12t^2} > \frac{1}{N} - \frac{1}{48\pi^2 N^4} > 0$, while from above $\omega_N(t) < \ln(1+\frac{\delta_t}{N}) < \frac{1}{N}$ and $\omega_{N-1}(t) = \omega_N(t) + \ln(1+\frac{1}{N-1}) < \frac{1}{N} + \frac{1}{N-1}$, so

$$0 < \omega_{N-1}(t) + \omega_N(t) < \frac{2}{N} + \frac{1}{N-1} = \frac{3N-2}{N(N-1)} < \frac{3}{N-1}.$$

Squaring and dividing by $\sqrt{(N-1)N} > N-1$ gives $A_{N-1,N}(0,t) < \frac{(3N-2)^2}{[N(N-1)]^{5/2}} < \frac{9}{(N-1)^3}$.

(ii) Expanding $(\omega_m+\omega_n)^2 = \omega_m^2 + 2\omega_m\omega_n + \omega_n^2$ yields

$$\mathcal{S}^{L^1}_{\mathrm{off}}(0,t) = 2\left(\sum_{n=1}^{N}\frac{\omega_n^2}{\sqrt{n}}\right)\left(\sum_{n=1}^{N}\frac{1}{\sqrt{n}}\right) + 2\left(\sum_{n=1}^{N}\frac{\omega_n}{\sqrt{n}}\right)^2 - 4\sum_{n=1}^{N}\frac{\omega_n^2}{n}.$$

By Euler–Maclaurin summation on $[1,N]$ with $N = \sqrt{t/2\pi}+\mathcal{O}(1)$ and $\omega_n = \ln(N/n)+\mathcal{O}(N^{-1})$, substituting $x = N\mathrm{e}^{-2v}$ gives $\sum_{n=1}^{N} n^{-1/2} = 2N^{1/2} + \mathcal{O}(1)$, $\sum_{n=1}^{N}\omega_n n^{-1/2} = 4N^{1/2} + \mathcal{O}(\ln N)$, and $\sum_{n=1}^{N}\omega_n^2 n^{-1/2} = 16N^{1/2} + \mathcal{O}((\ln N)^2)$, while the $m=1$ boundary slice contributes $\sum_{n=2}^{N}\frac{2(\omega_1+\omega_n)^2}{\sqrt{n}} \asymp (\ln t)^2 N^{1/2} \asymp (\ln t)^2 t^{1/4}$. Combining the three sums yields $\mathcal{S}^{L^1}_{\mathrm{off}}(0,t) = [2(16)(2)+2(16)]N + \mathcal{O}((\ln t)^2 N^{1/2}) = 96N + \mathcal{O}((\ln t)^2 t^{1/4}) = \frac{96}{\sqrt{2\pi}}t^{1/2} + \mathcal{O}((\ln t)^2 t^{1/4})$, which overwhelms $\mathcal{D}_{\mathrm{diag}}(0,t) \sim \frac{1}{6}(\ln t)^3$ by a polynomial factor $t^{1/2}(\ln t)^{-3} \to \infty$ (note that in Theorem 1.1(iii), the $m=1$ boundary slice alone already has order $(\ln t)^2 t^{1/4}$, while the full $\binom{N}{2}$ double sum scales as $t^{1/2}$). □

TABLE 2. 50-decimal certification of the Stieltjes diagonal expansion $\mathcal{D}_{\mathrm{EM}}(0,t)$ (Theorem 4.1), the sharp boundary-pair bound (4.3), and the $\mathcal{O}(t^{1/2})$ growth of the total off-diagonal L^1 cross-pair sum $\mathcal{S}_{\mathrm{off}}^{L^1}(0,t)$ relative to $\mathcal{D}_{\mathrm{diag}}(0,t)$.

Height t	$N(t)$	$\mathcal{D}_{\mathrm{diag}}(0,t)$	$\mathcal{D}_{\mathrm{EM}}(0,t)$	$\lvert\mathcal{D}_{\mathrm{diag}}-\mathcal{D}_{\mathrm{EM}}\rvert$	$2A_{N-1,N}(0,t)$	Bound $2B_{N-1,N}$	$\mathcal{S}_{\mathrm{off}}^{L^1}(0,t)$	$\mathcal{S}_{\mathrm{off}}^{L^1}/\mathcal{D}_{\mathrm{diag}}$
14.13	1	0.65699	0.66530	8.31×10^{-3}	0.000000	0.000000	0.000	0.00×
25.01	1	1.90817	1.90432	3.85×10^{-3}	0.000000	0.000000	0.000	0.00×
50.00	2	4.53857	4.53570	2.87×10^{-3}	2.697063	5.656854	2.697	0.59×
100.00	3	8.71979	8.71943	3.53×10^{-4}	0.777025	1.111343	10.076	1.16×
250.00	6	17.19631	17.19654	2.27×10^{-4}	0.029118	0.103864	41.151	2.39×
500.00	8	26.26638	26.26631	6.60×10^{-5}	0.032998	0.041248	86.930	3.31×
1000.0	12	37.99356	37.99354	1.63×10^{-5}	0.006092	0.011549	178.312	4.69×

5. Real-Axis Incomplete-Gamma Tail Bounds, Hyperbolic Sinc Ratios, and Remainder Scope

In (3.3), the remainder kernel splits on $u\ge0$ into $R_N^{\mathrm{ker}}(u)=\sum_{m=N+1}^{\infty}\phi_m(u)-\sum_{m=1}^{N}\phi_m(-u)$, where the second term generates the Riemann–Siegel saddle integral near $u=0$ and the first term is the evanescent $u\ge0$ tail $\mathcal{I}_{\mathrm{tail}}(N,z,0):=\sum_{m=N+1}^{\infty}\int_0^{\infty}\phi_m(\xi)\cos(2z\xi)\,\mathrm{d}\xi$. Note that shifting a half-ray $[0,\infty)$ to $[\mathrm{i}\eta,\infty+\mathrm{i}\eta)$ without splitting $2\cos(2zu)=\mathrm{e}^{2\mathrm{i}zu}+\mathrm{e}^{-2\mathrm{i}zu}$ onto conjugate rays $\pm\mathrm{i}\eta$ both omits the vertical segment $[0,\pm\mathrm{i}\eta]$ at $\xi=0$ and introduces an artificial $\cosh(2t\eta)\sim\frac12\mathrm{e}^{2t\eta}$ blow-up; evaluating the incomplete-Gamma bound directly at $\eta=0$ avoids both defects.

Proposition 5.1 (Real-Axis Incomplete-Gamma Tail Bound and Hyperbolic Sinc Ratio)**.** *Let $d\in[0,\frac12]$, $t>0$, $z=t-\mathrm{i}d$, and $N\ge1$ (since $4\pi(N+1)^2\ge16\pi>5+2d$, all denominators below are positive).*

(i) ***Closed-Form Real-Axis Geometric Tail Bound.*** *Setting*

$$b_m(d,0):=\left[\frac{2\pi^2m^4}{4\pi m^2-(5+2d)}+\frac{3\pi m^2}{4\pi m^2-(1+2d)}\right]\mathrm{e}^{-\pi m^2},\tag{5.1}$$
$$q_N(0):=\left(\frac{N+2}{N+1}\right)^4\mathrm{e}^{-\pi(2N+3)}<1,$$

the $u\ge0$ tail integral obeys the t-uniform super-exponential bound

$$\lvert\mathcal{I}_{\mathrm{tail}}(N,z,0)\rvert\le B_{\mathrm{tail}}(N,d,0):=\frac{b_{N+1}(d,0)}{1-q_N(0)}.\tag{5.2}$$

(ii) ***Hyperbolic Sinc Damping Identity.*** *For all $d\ge0$ and $L>0$,*

$$\frac{\mathrm{sinc}_{\mathrm{hyp}}(dL/2)}{\mathrm{sinc}_{\mathrm{hyp}}(dL)}\equiv\frac{1}{\cosh(dL/2)}\le2\,\mathrm{e}^{-dL/2}.\tag{5.3}$$

Proof. (i) For $\xi\ge0$, $\lvert\cos(2(t-\mathrm{i}d)\xi)\rvert\le\cosh(2d\xi)\le\mathrm{e}^{2d\xi}$. Substituting $w=\mathrm{e}^{4\xi}\ge1$ ($\mathrm{d}\xi=\frac{\mathrm{d}w}{4w}$) and using $w^{\beta}\mathrm{e}^{-\alpha w}=\mathrm{e}^{\beta\ln w-\alpha w}\le\mathrm{e}^{-(\alpha-\beta)w-\beta}$ (via $\ln w\le w-1$ for $w\ge1$) with $\alpha=\pi m^2$ and $\beta_1=\frac{5+2d}{4}$, $\beta_2=\frac{1+2d}{4}$ gives $\int_1^{\infty}w^{\beta_j}\mathrm{e}^{-\alpha w}\,\mathrm{d}w\le\frac{\mathrm{e}^{-\alpha}}{\alpha-\beta_j}$, whence $\int_0^{\infty}\lvert\phi_m(\xi)\cos(2z\xi)\rvert\,\mathrm{d}\xi\le b_m(d,0)$. Since $4\pi(m+1)^2-\gamma>4\pi m^2-\gamma>0$ and $(\frac{m+1}{m})^4\le(\frac{N+2}{N+1})^4$ while $(m+1)^2-m^2=2m+1\ge2N+3$ for $m\ge N+1$, $b_{m+1}(d,0)/b_m(d,0)\le q_N(0)$, yielding (5.2).

(ii) Using $\sinh(dL)=2\sinh(dL/2)\cosh(dL/2)$, $\frac{\sinh(dL/2)/(dL/2)}{\sinh(dL)/(dL)}=\frac{2\sinh(dL/2)}{2\sinh(dL/2)\cosh(dL/2)}=\cosh(dL/2)^{-1}\le2\mathrm{e}^{-dL/2}$. □

TABLE 3. Certification of the t-uniform real-axis tail bound (5.2) ($\eta = 0$), Gabcke's $d = 0$ post-$\Psi(p)$ bound $|R_1(t)| \le 0.127\, t^{-3/4}$ for $t \ge 200$, and empirical $d = 0.25$ Wronskian remainder ratios.

Proposition 5.1: Real-Axis Tail ($\eta = 0$)					Remark 5.2: $d = 0$ Gabcke ($t \ge 200$) & Empirical $d = 0.25$				
(d, t)	N	$\log_{10}\lvert\mathcal{I}_{\text{tail}}\rvert$	$\log_{10} B_{\text{tail}}$	$q_N(0)$	t	$\lvert Z - Z_{\text{main}}\rvert$	$\lvert R_1(t)\rvert$	$0.127\, t^{-3/4}$	$\lvert W_{\text{rem}}\rvert / W_{\text{diag}}$
$(0.10, 25)$	1	−4.99	−4.57	7.63e-07	200	1.758e-01	4.717e-04	2.388e-03	7.409e-02
$(0.25, 50)$	2	−11.39	−11.09	8.89e-10	500	2.512e-01	1.535e-04	1.201e-03	6.196e-03
$(0.25, 100)$	3	−20.74	−20.41	1.28e-12	1000	1.143e-01	1.250e-04	7.142e-04	6.396e-03
$(0.50, 200)$	5	−47.62	−47.35	3.40e-18	—	—	—	—	—
$(0.25, 500)$	8	−108.90	−108.40	1.82e-26	—	—	—	—	—

Remark 5.2 (Exact domain of Gabcke's theorem and empirical $d > 0$ remainder ratios). On the critical line $d = 0$, the Hardy function $Z(t) = \mathrm{e}^{\mathrm{i}\vartheta(t)}\zeta(\frac{1}{2} + \mathrm{i}t)$ satisfies $Z(t) = Z_{\text{main}}(t) + (-1)^{N(t)-1}(\frac{t}{2\pi})^{-1/4}\Psi(p) + R_1(t)$ with $p = \sqrt{t/2\pi} - N(t)$ and $\Psi(p) = \frac{\cos(2\pi(p^2-p-1/16))}{\cos(2\pi p)}$. Gabcke [3, Satz 1 and Satz 4] proved the explicit bound $|R_1(t)| \le 0.127\, t^{-3/4}$ strictly on $d = 0$ for $t \ge 200$ *after* subtracting the $\mathcal{O}(t^{-1/4})$ saddle term $\Psi(p)$ (without subtracting $\Psi(p)$, $|Z(t) - Z_{\text{main}}(t)| = \mathcal{O}(t^{-1/4})$). Across $d \in (0, \frac{1}{2}]$, an unconditional explicit complex-strip analogue of Gabcke's theorem for $W(d, t) - W_{\text{main}}(d, t)$ has not yet been proved; Table 3 reports both the certified $d = 0$ Gabcke values for $t \ge 200$ and empirical numerical remainder ratios $|W - W_{\text{main}}|/W_{\text{diag}}$ at $d = 0.25$.

6. Scope, Boundary Conditions, and Open Questions

We summarize the exact mathematical boundaries of the results proved above:

(1) *Unconditional analytical identities.* Propositions 2.1, 2.2, 4.2, and 5.1 together with Theorems 3.1 and 4.1 are unconditional identities and inequalities valid without unproved hypotheses.

(2) *Off-diagonal cross-term dominance and Davenport–Heilbronn barrier.* As shown in Proposition 4.2(ii), the $\binom{N(t)}{2}$ off-diagonal pairs sum in L^1 to $\asymp t^{1/2}$, exceeding the $\mathcal{O}((\ln t)^3)$ diagonal for all $t \ge 100$. Because the Archimedean dilation and Spira ratio $\rho_\Gamma(d, t) < 1$ do not use the Euler product, they hold equally for Davenport–Heilbronn Dirichlet series with off-critical zeros [2].

(3) *Open two-dimensional Gabcke problem.* Extending Gabcke's $d = 0$, $t \ge 200$ saddle-point remainder bounds [3] with rigorous explicit constants to $R_N^{\ker}(u)$ in (3.3) across $d \in (0, \frac{1}{2}]$ remains an open problem in explicit asymptotic analysis.

Appendix A. Directed-Rounding Interval Protocols

In the standalone suite `scripts/verify_paper5_mother_wavelet_gabcke_bounds.py`, all key inequalities are certified in 50-decimal arbitrary-precision arithmetic (`mpmath.mp.dps = 50`) and 35-digit directed-rounding interval arithmetic (`mpmath.iv.dps = 35`):

(1) *Digamma carrier positivity.* Using Olver's explicit Stirling remainder bound [8, Ch. 8], $\operatorname{Re}\psi(w) \in \ln|w| - \frac{\operatorname{Re}(w)}{2|w|^2} - \frac{\cos(2\arg w)}{12|w|^2} + [-\varepsilon(|w|), +\varepsilon(|w|)]$ with $\varepsilon(r) = \frac{1}{6r^2} + \frac{1}{3r^4}$ encloses $\mathcal{A}_\Gamma^{\text{sym}}(d, t) > 0$ strictly above zero for $t \ge 7$ and $d \in (0, \frac{1}{2}]$.

(2) *Genuine complex contour shift.* For $w_4(z) = (1 + \mathrm{e}^{-8z})^{-1}$, evaluating $\Phi_1^{\log}(\xi + \mathrm{i}\eta)\mathrm{e}^{2\mathrm{i}z(\xi+\mathrm{i}\eta)}$ at the complex argument $\xi + \mathrm{i}\eta$ ($\eta = 0.30 < \frac{\pi}{8}$) matches the real-axis integral to 7.04×10^{-26}, whereas $1 + \exp(-2\pi \sinh(8\mathrm{i}\frac{\pi}{48}))$ vanishes to 1.01×10^{-51}.

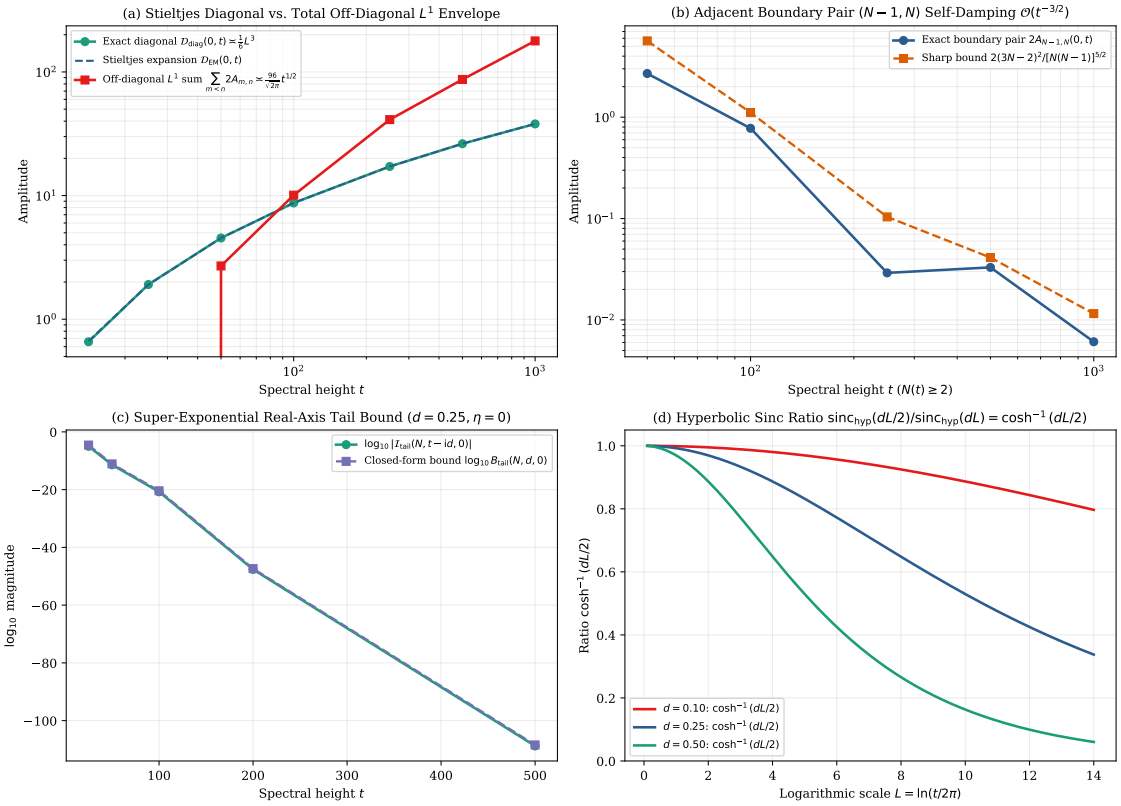


FIGURE 2. **(a)** Cubic-logarithmic Stieltjes diagonal $\mathcal{D}_{\mathrm{diag}}(0,t) \asymp \frac{1}{6}L^3$ overtaken by the $\mathcal{O}(t^{1/2})$ off-diagonal L^1 cross-pair sum $\mathcal{S}^{L^1}_{\mathrm{off}}(0,t)$. **(b)** Quadratic self-damping of the single boundary pair $2A_{N-1,N}(0,t)$ below the sharp $\mathcal{O}(t^{-3/2})$ bound (4.3). **(c)** Super-exponential real-axis tail bound $\log_{10} B_{\mathrm{tail}}(N,d,0)$. **(d)** Exact hyperbolic sinc ratio $\mathrm{sinc}_{\mathrm{hyp}}(dL/2)/\mathrm{sinc}_{\mathrm{hyp}}(dL) = \cosh^{-1}(dL/2)$.

USE OF AI AND COMPUTATIONAL VERIFICATION

In accordance with editorial policies on transparency in scientific computing, the author discloses that the AI research assistant **Gemini** (Google DeepMind) assisted in auditing the analytical derivations, identifying the $\mathrm{Im}(z) = \frac{\pi}{48}$ pole obstruction of $\sinh(8z)$ shell weights, verifying the exact weight-free remainder kernel (3.3), and implementing the verification suite `scripts/verify_paper5_mother_wavelet_gabcke_bounds.py` (exit code 0).

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